

Notes: Andries, Bianchi, Huynh & Pouget (RFS, 2025)
Return Predictability, Expectations, and Investment: Experimental Evidence

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1 Big picture

- **Question.** How do investors form beliefs about returns, and how do those beliefs translate into risky portfolio allocations when the information environment changes?
- **Core design.** Subjects repeatedly see time-series graphs of an “Index Return” and a signal called “Variable A.” In some rounds Variable A predicts the next return; in other rounds it is useless.
- **Information-beliefs-decisions chain.** Each round elicits whether Variable A is useful, the subject’s forecast, and the risky investment out of 100 ECU.
- **Main belief result.** When subjects perceive Variable A as useful, they use it and stop extrapolating from past returns. When they perceive Variable A as useless, they extrapolate.
- **Main decision result.** Forecasts affect investment, but the pass-through is stronger for signal-based forecasts than for extrapolative forecasts.
- **Interpretation.** The same investor can behave like an extrapolative household investor or a more rational asset manager depending on the information environment.

2 Data and measurement

2.1 Experimental environment

Subjects observe 40-period histories of Index Return and Variable A. The Index Return mimics U.S. equity index five-year returns and has no serial autocorrelation. Variable A has the same unconditional distribution in all rounds, but it predicts returns in some rounds and is useless in others.

Interpretation: The design creates controlled uncertainty about whether the available signal should be used.

2.2 Return processes

$$\text{i.i.d. rounds: } r_{t+1} = \mu + \varepsilon_{t+1}, \quad \text{predictable rounds: } r_{t+1}^p = a_t + \varepsilon_{t+1}^p.$$

Variable A is calibrated to a dividend-price-ratio-like predictive signal. Subjects are not told the formula.

Interpretation: The signal is realistic: potentially useful, observable, and interpretable only through data.

2.3 Outcomes

The main variables are $Predict_{i,k}$, the subject’s perceived usefulness of Variable A; $F_{i,k}$, the next-period forecast; $\theta_{i,k}$, the risky investment; confidence intervals; and upper/lower tail probabilities.

Interpretation: The paper observes information perception, beliefs, and decisions at the same time.

3 Models and empirical specifications

3.1 Forecast regression

$$F_{i,k} = \alpha_1 + \alpha_2 Predict_{i,k} + \beta_1 a_{t,k} + \beta_2 a_{t,k} Predict_{i,k} + \delta_1 r_{t,k} + \delta_2 r_{t,k} Predict_{i,k} + \epsilon_{i,k}.$$

Interpretation: This equation tests whether subjects use Variable A only when they think it is useful and extrapolate only when they do not.

3.2 Investment regression

$$\theta_{i,k} = \alpha_1 + \alpha_2 Predict_{i,k} + \beta_1 F_{i,k} + \beta_2 F_{i,k} Predict_{i,k} + \epsilon_{i,k}.$$

Interpretation: The interaction coefficient measures whether signal-based forecasts receive more weight in portfolio choice.

3.3 Informed forecasts and 2SLS

$$\theta_{i,k} = \alpha + \beta F_{i,k}^{informed} + \epsilon_{i,k},$$

where forecasts are instrumented by a_t in Predict=1 rounds and by r_t in Predict=0 rounds.

Interpretation: This isolates the forecast component generated by the information source the subject appears to use.

3.4 Risk regressions

$$\begin{aligned} \theta_{i,k} &= \alpha + \beta F_{i,k} + \gamma HighCI_{i,k} + \eta F_{i,k} HighCI_{i,k} + \epsilon_{i,k}. \\ \theta_{i,k} &= \alpha + \beta F_{i,k} + \gamma_H HighProbHigh_{i,k} + \gamma_L HighProbLow_{i,k} + \dots + \epsilon_{i,k}. \end{aligned}$$

Interpretation: These regressions separate mean forecasts, second-moment uncertainty, and tail-probability/skewness channels.

3.5 Cognitive uncertainty model

$$\theta_i = \frac{1}{\gamma_i} \frac{E_i(r)}{\sigma_i^2(r)}, \quad E_{i,k}(r) = \xi_i E(F_i) + (1 - \xi_i) F_{i,k}^e.$$

Interpretation: The parameter ξ_i captures stickiness of effective beliefs around average forecasts. Larger ξ_i implies weaker decision response to forecast changes.

4 Identification and experimental design

The experiment identifies mechanisms using within-subject variation in information environments. Individual fixed effects absorb stable preferences and ability; round fixed effects absorb common graph realizations. The key innovation is observing the subject's perceived information set, forecast, and investment jointly. The design cannot fully reproduce market equilibrium, transaction costs, or multi-asset portfolios, but it cleanly isolates the information-beliefs-decisions chain.

5 Section-by-section study guide

- **Introduction:** Explains why field data cannot usually observe information, beliefs, and decisions together.
- **Section 1:** Describes the experiment, incentives, return processes, and subject waves.
- **Section 2:** Provides the core empirical evidence in Tables 1–6.
- **Section 3:** Interprets the evidence using a dual-expectation model and cognitive uncertainty.
- **Section 4:** Shows external-validity and heterogeneity results.
- **Section 5:** Explains why partial observation leads to wrong conclusions and discusses market implications.

6 Tables and figures

6.1 Figure 1: Example page

Read: The figure shows the graphical screen subjects see: Index Return, Variable A, and the last realization of Variable A. One example is predictive and one is not.

Detailed interpretation: This figure is central because subjects infer the usefulness of the signal from visual time-series information rather than from a formula.

Economic message: The information environment is the treatment: perceived signal usefulness drives belief formation and decision sensitivity.

6.2 Table 1: Descriptive statistics

Read: Subjects identify predictive rounds 82% of the time and useless rounds 70% of the time. Forecasts are more accurate and investments are higher when Variable A is perceived predictive.

Detailed interpretation: Table 1 shows partial but imperfect learning: subjects understand the signal often enough for the treatment to matter, but errors remain realistic.

Economic message: Useful perceived information raises both forecast accuracy and risk-taking.

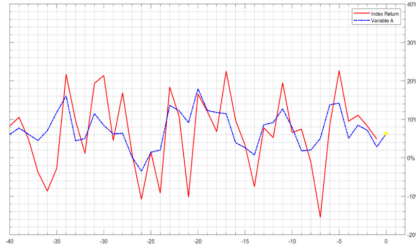
6.3 Table 2: Forecast and predictability

Read: Variable A affects forecasts only when perceived useful; past returns affect forecasts mainly when Variable A is not perceived useful.

Detailed interpretation: The table rejects a single forecast model. Subjects switch between signal-based and extrapolative forecasting rules.

Economic message: Useful signals crowd out extrapolation.

Below are examples of what you may see during the experiment:



The red line represents the past *returns* of the index from period -40 to period -1.

The blue dashed line is the past realizations of Variable A, from period -40 to 0. Today's value of Variable A is indicated by the yellow dot. In certain rounds, this yellow dot can be useful for predicting the index returns. We are at date 0, today.

In the graph above, Variable A is useful to predict the index returns.

Here is another example of the graphs that you may see. In this second graph, Variable A is not useful in predicting the index returns.

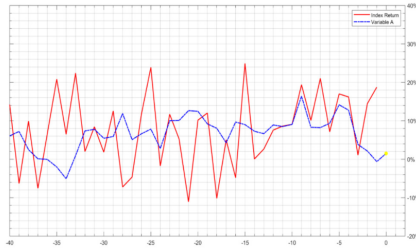


Table 1
Descriptive statistics

Variable	Obs.	Mean	Median	Std. dev.	Min	Max
$Pr(A \text{ perceived predictive} \text{predictable})$	169	0.82	0.80	0.14	0.30	1
$Pr(A \text{ perceived useless} \text{i.i.d.})$	169	0.70	0.70	0.20	0.20	1
Predict	169	0.56	0.55	0.14	0.20	0.90
Forecast (in %)	3,380	5.9	6	8.0	-30	100
Forecast distance (in %)	3,380	9.0	7.2	8.0	0.0	93.8
Invest (in ECU)	3,380	42.6	35	36.0	0	100
5-year forecast (in %)	1,080	6.7	6	7.7	-15	100
5-year invest (in ECU)	1,080	52.4	50	33.4	0	100
Predict=1						
Forecast (in %)	1,888	6.5	7	7.6	-30	70
Confidence interval (in %)	393	20.4	20	14.5	1	88
Upper prob. (in %)	660	22.3	15	23.0	0	100
Lower prob. (in %)	660	20.3	10	22.9	0	100
Forecast distance (in %)	1,888	7.7	6.2	6.5	0.0	78.0
Invest (in ECU)	1,888	46.4	40	36.3	0	100
5-year forecast (in %)	566	7.4	6	8.9	-15	100
5-year invest (in ECU)	566	53.9	50	33.9	0	100
Predict=0						
Forecast (in %)	1,492	5.1	5	8.4	-20	100
Confidence interval (in %)	287	21.1	20	14.1	1	82
Upper prob. (in %)	480	19.4	10	21.1	0	100
Lower prob. (in %)	480	28.3	20	25.1	0	100
Forecast distance (in %)	1,492	10.7	8.5	9.4	0.1	93.8
Invest (in ECU)	1,492	37.9	25	35.0	0	100
5-year forecast (in %)	514	6.1	6	6.0	-14.5	80
5-year invest (in ECU)	514	50.7	50	32.8	0	100
"Variable A" is revealed predictive						
Forecast (in %)	460	6.3	7	6.4	-15	28
Invest (in ECU)	460	54.1	50	38.4	0	100
"Variable A" is revealed not predictive						
Forecast (in %)	460	6.0	6	7.2	-15	30
Invest (in ECU)	460	49.1	50	38.5	0	100
"Model revealed" treatment - Predict = 1						
Forecast (in %)	144	5.8	6.3	5.6	-15	20
Invest (in ECU)	144	55.0	50	37.6	0	100
"Model revealed" treatment - Predict = 0						
Forecast (in %)	94	4.7	5	5.7	-16	17
Invest (in ECU)	94	43.0	30	39.9	0	100

"Predict" is a dummy equal to one if the subject perceives "Variable A" is useful to predict returns. "Predict=1" and "Predict=0" results correspond to rounds perceived as predictable or not by "Variable A" in the baseline treatment across all waves.

the "Variable A" information as useful may also reflect optimism bias (here, a

Table 2
Forecast and predictability

Dep variable	Forecast								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
a(t)	0.24*** (0.05)		0.06 (0.07)	0.01 (0.06)	0.01 (0.06)				0.15** (0.06)
a(t) × Predict			0.28*** (0.10)	0.36*** (0.08)	0.37*** (0.08)				0.30*** (0.07)
r(t)		0.10 (0.03)				0.18*** (0.03)	0.18*** (0.03)	0.18*** (0.03)	0.19*** (0.03)
r(t) × Predict						-0.17*** (0.04)	-0.17*** (0.04)	-0.17*** (0.04)	-0.12*** (0.04)
Predict			-0.47 (0.53)	-0.90* (0.47)	-0.95* (0.47)	1.71*** (0.34)	1.71*** (0.40)	1.73*** (0.40)	-0.42 (0.50)
N	3,380	3,380	3,380	3,380	3,380	3,380	3,380	3,380	3,380
R ²	.15	.15	.02	.15	.16	.03	.16	.16	.18
Individual FE	Yes	Yes	No	Yes	Yes	No	Yes	Yes	Yes
Round FE	Yes	Yes	No	No	Yes	No	No	Yes	Yes

This table reports the results of OLS regressions. The dependent variable is the forecast of next period returns in percentage points. "Predict" is a dummy equal to one if the subject declares "Variable A" is useful to predict returns. a(t) denotes the last realization of "Variable A." r(t) denotes the last realization of "Index Return." Two-way clustered standard errors (round and individual levels) are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 3
Investment and forecasts

Dep variable	Investment					
	(1)	(2)	(3)	(4)	(5)	(6)
Forecast	1.60*** (0.25)	1.67*** (0.18)	1.67*** (0.19)	1.27*** (0.24)	1.36*** (0.19)	1.38*** (0.20)
Forecast × Predict				0.58*** (0.08)	0.52*** (0.14)	0.48*** (0.13)
Predict				3.12** (1.34)	4.07*** (0.91)	4.33*** (0.91)
N	3,380	3,380	3,380	3,380	3,380	3,380
R ²	.13	.55	.58	.14	.56	.59
Individual FE	No	Yes	Yes	No	Yes	Yes
Round FE	No	No	Yes	No	No	Yes

This table reports the results of OLS regressions. The dependent variable is the endowment invested in the risky asset, in ECU. "Forecast" is the forecast of next period returns in percentage points. "Predict" is a dummy equal to one if the subject declares that "Variable A" is useful to predict returns. Two-way clustered standard errors (round and individual levels) are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

6.4 Table 3: Investment and forecasts

Read: Forecasts strongly predict risky investment. Forecast $\times$ Predict is positive and significant.

Detailed interpretation: The same forecast receives greater decision weight when it is supported by perceived predictive information.

Economic message: Belief-to-decision pass-through depends on information source.

6.5 Table 4: Investment and “informed forecasts”

Read: Instrumenting forecasts by the relevant information source yields a 2SLS pass-through of 3.19 in Predict=1 rounds and 1.56 in Predict=0 rounds.

Detailed interpretation: Signal-based forecasts influence investments about twice as strongly as extrapolative forecasts.

Economic message: Investors trust predictive-signal forecasts more than extrapolative forecasts.

Table 4
Investment and “informed forecasts”

Dep variable	Investment			
	(1) 2SLS	(2) OLS	(3) 2SLS	(4) OLS
Forecast	3.19*** (0.67)	1.85*** (0.10)	1.56*** (0.29)	1.43*** (0.14)
N	1,888	1,888	1,492	1,492
Sample	Predict=1		Predict=0	
Instrument	a(t)		r(t)	
Individual FE	Yes	Yes	Yes	Yes
Round FE	Yes	Yes	Yes	Yes

This table reports the results of the 2SLS regressions (5), and the OLS regression of Equation (4). The dependent variable is the endowment invested in the risky asset, in ECU. “Forecast” is the forecast of next period returns in percentage points. “Predict” is a dummy equal to one if the subject declares that “Variable A” is useful to predict returns. In the 2SLS columns, “Forecast” is instrumented by a_t , the last realization of “Variable A,” when “Predict=1,” and by r_t , the last realization of “Index Return,” when “Predict=0.” Clustered standard errors, at the round level, are in parentheses (computing standard errors clustered at the individual-round level would yield a singular covariance matrix). * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 5
Investment and confidence intervals

Dep variable	Investment					
	(1)	(2)	(3)	(4)	(5)	(6)
Forecast	2.81*** (0.40)	2.42*** (0.42)	2.33*** (0.42)	2.46*** (0.40)	2.19*** (0.60)	2.13*** (0.61)
High CI	1.54 (2.50)	1.14 (2.82)	−0.79 (2.34)	6.72 (4.96)	2.84 (2.92)	1.95 (3.52)
Forecast $\times$ High CI	−0.26 (0.42)	−0.10 (0.48)	−0.08 (0.44)	−1.65*** (0.28)	−0.97 (0.57)	−0.85 (0.57)
N	393	393	393	287	287	287
R ²	.25	.63	.68	.08	.66	.68
Sample	Predict=1			Predict=0		
Individual FE	No	Yes	Yes	No	Yes	Yes
Round FE	No	No	Yes	No	No	Yes

This table reports the results of OLS regressions. The dependent variable is the fraction of the endowment invested in the risky asset, in percentage points. “Forecast” is the forecast of next period returns in percentage points. “High CI” is a dummy equal to one in rounds where the reported confidence interval is above or equal the subject’s median value for the same round type – perceived as predictable or not by “Variable A.” “Predict” is a dummy equal to one if the subject declares that “Variable A” is useful to predict returns. Two-way clustered standard errors (round and individual levels) are in parentheses. These results were obtained during wave two of the experiment implementation (TSE lab, January 2020). * $p < .1$; ** $p < .05$; *** $p < .01$.

6.6 Table 5: Investment and confidence intervals

Read: Forecasts remain important, but confidence intervals do not systematically change investment or forecast pass-through.

Detailed interpretation: The low investment elasticity is not primarily a standard second-moment uncertainty effect.

Economic message: Reported variance uncertainty is not the main explanation for underreaction.

6.7 Table 6: Investment and upper/lower-bound probabilities

Read: High upside-tail probability raises investment; high downside-tail probability lowers investment.

Detailed interpretation: The opposite signs are consistent with skewness preferences or tail distortions, not just higher perceived variance.

Economic message: Demand responds to perceived tail shape, not only expected return.

Table 6
Investment and upper/lower-bound probabilities

Dep variable	Investment					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A</i>						
Forecast	1.37*** (0.36)	1.52*** (0.34)	1.54*** (0.31)	0.86** (0.34)	1.28** (0.48)	1.18** (0.44)
HighProbHigh	9.83** (4.45)	9.62** (3.37)	10.14*** (3.17)	5.93 (3.59)	10.84*** (3.74)	10.41** (3.76)
Forecast $\times$ HighProbHigh	-0.51 (0.36)	-0.61** (0.23)	-0.67*** (0.18)	0.11 (0.27)	-0.49 (0.42)	-0.31 (0.37)
N	660	660	660	480	480	480
R^2	.09	.58	.64	.10	.61	.64
Sample	Predict=1			Predict=0		
<i>Panel B</i>						
Forecast	0.43 (0.39)	0.61* (0.32)	0.58 (0.40)	0.63* (0.35)	0.86*** (0.26)	0.86*** (0.25)
HighProbLow	-15.18*** (4.41)	-15.45*** (4.09)	-15.44*** (4.35)	-7.40* (3.83)	-7.69** (3.01)	-8.06** (3.02)
Forecast $\times$ HighProbLow	0.91* (0.45)	0.74* (0.35)	0.77* (0.40)	0.51 (0.37)	0.03 (0.27)	0.11 (0.26)
N	660	660	660	480	480	480
R^2	.09	.59	.64	.10	.61	.64
Sample	Predict=1			Predict=0		
<i>Panel C</i>						
Forecast	0.53 (0.60)	0.87 (0.54)	0.88 (0.56)	0.44 (0.43)	1.14** (0.52)	0.93* (0.47)
HighProbHigh	8.04* (4.48)	8.21** (3.60)	8.74** (3.27)	5.51 (3.60)	10.47*** (3.62)	10.02** (3.62)
HighProbLow	-14.27** (5.12)	-13.86*** (4.12)	-13.73*** (4.34)	-7.25* (3.68)	-6.76** (2.95)	-7.65** (2.97)
Forecast $\times$ HighProbHigh	-0.23 (0.41)	-0.43 (0.32)	-0.48* (0.27)	0.17 (0.30)	-0.46 (0.41)	-0.24 (0.35)
Forecast $\times$ HighProbLow	0.81 (0.51)	0.56 (0.41)	0.57 (0.44)	0.52 (0.38)	-0.04 (0.26)	0.10 (0.26)
N	660	660	660	480	480	480
R^2	.10	.60	.65	.10	.62	.65
Sample	Predict=1			Predict=0		
Individual FE	No	Yes	Yes	No	Yes	Yes
Round FE	No	No	Yes	No	No	Yes

This table reports the results of OLS regressions. The dependent variable is the fraction of the endowment invested in the risky asset, in percentage points. "Forecast" is the forecast of next period returns in percentage points. "HighProbHigh" is a dummy equal to one in rounds where the reported upper-bound probability (probability that next period return is above 15%) is above or equal the subject's median value for the same round type – perceived as predictable or not by "Variable A." "HighProbLow" is a dummy equal to one in rounds where the reported lower-bound probability (probability that next period return is below -3%) is above or equal the subject's median value for the same round type – perceived as predictable or not by "Variable A." "Predict" is a dummy equal to one if the subject declares that "Variable A" is useful to predict returns. Two-way clustered standard errors (round and individual levels) are in parentheses. These results were obtained during wave four of the experiment implementation (Prolific online, July 2023). * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 7
Forecast model

	Model (1)	Data (2)	Difference (3)	p -value (4)
α_1	3.79 [3.50, 4.08]	3.83 [3.18, 4.48]	-0.04 [-0.75, 0.67]	.911
α_2	-1.36 [-1.56, -1.15]	-0.42 [-1.40, 0.56]	-0.94 [-1.94, 0.07]	.067
β_1	0.11	0.15 [0.03, 0.26]	-0.03 [-0.14, 0.08]	.599
β_2	0.41	0.30 [0.17, 0.43]	0.11 [-0.02, 0.24]	.108
δ_1	0.26 [0.21, 0.31]	0.19 [0.13, 0.26]	0.07 [-0.01, 0.15]	.103
δ_2	-0.18 [-0.22, -0.15]	-0.12 [-0.19, -0.05]	-0.06 [-0.14, 0.02]	.129

In column 1, we report the average predicted values according to the forecast model of Section 3.1. The confidence intervals are obtained by plugging the upper-bound and lower-bound values of the extrapolation coefficient estimated by [Lander, Ma, and Thesmar \(2019\)](#); [Afrouzi et al. \(2023\)](#). $\epsilon_u = 0.32$ with S.E. 0.03, into the forecast model of Section 3.1. In column 1 we make the conservative assumption that the probabilities of mistakes when identifying "Variable A" are estimated without errors, for each subject. In column 2, we report the estimates of the OLS regression (3): $F_{i,k} = \alpha_1 + \alpha_2 \text{Predict}_{i,k} + \beta_1 a_{i,k} + \beta_2 a_{i,k} \times \text{Predict}_{i,k} + \delta_1 r_{i,k} + \delta_2 r_{i,k} \times \text{Predict}_{i,k} + \epsilon_{i,k}$, estimated with round and subject fixed effects. The confidence intervals are obtained using standard errors two-way clustered by round and subject. The 95% confidence interval in column 3 are estimated using standard errors that are computed as $\sqrt{\sigma_m^2 + \sigma_d^2}$, where σ_m^2 and σ_d^2 are the standard errors as in columns 1 and 2, respectively. In column 4, we report the p -values of the t -tests that the difference in column 3 is equal to zero.

6.8 Table 7: Forecast model

Read: The dual-expectation model matches the main forecast coefficients and average levels reasonably well.

Detailed interpretation: The model combines extrapolation, signal updating, and mistakes in identifying Variable A.

Economic message: The belief model quantitatively disciplines the forecast patterns.

6.9 Table 8: Forecast and investment, TSE students versus Prolific subjects

Read: Both subject pools display the same qualitative forecast and investment mechanisms.

Detailed interpretation: Subject pool differences mostly change magnitudes, not the structure of the information-beliefs-decisions chain.

Economic message: The mechanism is not an artifact of finance-student participants.

Table 8
Forecast and investment, TSE students versus Prolific subjects

Dep variable	Forecast		Investment	
	(1)	(2)	(3)	(4)
a(t)	-0.03 (0.07)			
a(t) × Prolific	0.14 (0.13)			
a(t) × Predict	0.42*** (0.09)			
a(t) × Predict × Prolific	-0.16 (0.18)			
r(t)		0.21*** (0.04)		
r(t) × Prolific		-0.09 (0.08)		
r(t) × Predict		-0.20*** (0.05)		
r(t) × Predict × Prolific		0.07 (0.10)		
Forecast			2.03*** (0.17)	1.69*** (0.22)
Forecast × Prolific			-0.86** (0.30)	-0.68* (0.33)
Forecast × Predict			0.54** (0.20)	0.54** (0.20)
Forecast × Predict × Prolific			-0.26 (0.30)	-0.26 (0.30)
Predict	-1.12* (0.57)	1.92*** (0.40)		3.47** (1.30)
Predict × Prolific	0.56 (1.16)	-0.56 (0.78)		2.76 (2.74)
N	3,380	3,380	3,380	3,380
R ²	.16	.16	.59	.60
Individual FE	Yes	Yes	Yes	Yes
Round FE	Yes	Yes	Yes	Yes

This table reports the results of OLS regressions. In columns 1 and 2, the dependent variable is the next-period forecast of returns, in percentage points. In columns 3 and 4, the dependent variable is the ECU next-period investment in the risky asset. "Predict" is a dummy equal to one if the subject declares "Variable A" is useful to predict returns. "Prolific" is a dummy equal to one if the subject was recruited via the online Prolific platform. Two-way clustered standard errors (round and individual levels) are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 9
Average ξ across subsamples

Subsample	"Variable A" perceived useful	"Variable A" perceived useless
High investments	0.58	0.69
Low investments	0.37	0.78
Fast	0.36	0.58
Slow	0.56	0.81
High ability	0.47	0.75
Low ability	0.52	0.74
Female	0.50	0.60
Male	0.49	0.88
High grades	0.51	0.68
Low grades	0.45	0.87
Young	0.58	0.64
Old	0.47	0.75
High education	0.63	0.59
Low education	0.99	0.72
High income	0.65	0.58
Low income	0.79	0.69
High fin. literacy	0.48	0.71
Low fin. literacy	0.83	0.44
All	0.50	0.74

This table reports the average of ξ estimated on samples of subjects, using the elasticity to "informed forecasts" $\{\bar{F}_{i,t}\}$ in both round types. "High θ " is a dummy equal to one if the subject takes larger risk investments, on average, than the median; "Fast" is a dummy equal to one if the subject is faster, on average, than the median seconds in answering each round's questions; "High ability" is a dummy equal to one if the subject is better than the median in identifying when "Variable A" is useful or not; "High grades" is a dummy equal to one if the subject has average grades above her/his cohort's median in TSE Master's program; "Female" is a dummy equal to one if the subject is a woman. The variables "Young," "High education," "High income," and "High fin. literacy" only apply to Prolific subjects: "Young" is a dummy equal to one if the subject's age is less than the median of all Prolific subjects. "High education" is a dummy equal to one if the subject has a 4-year college degree. "High income" is dummy equal to one if the subject has annual income above \$50,000. "High fin. literacy" is a dummy equal to one if the subjects answer three financial literacy questions correctly.

6.10 Table 9: Average ξ across subsamples

Read: Cognitive uncertainty ξ is higher when Variable A is perceived useless than when it is perceived useful.

Detailed interpretation: Most subgroups show higher stickiness for extrapolative forecasts, supporting the source-confidence interpretation.

Economic message: Extrapolative beliefs are less actionable than signal-based beliefs.

7 Short summary

- Subjects use Variable A in forecasts only when they perceive it as predictive.
- Otherwise, they extrapolate from past returns.
- Forecasts affect investments, but the pass-through is too low relative to Merton-Samuelson.
- Signal-based forecasts have stronger pass-through than extrapolative forecasts.
- Confidence intervals do not explain the low pass-through, while tail beliefs matter independently.
- Cognitive uncertainty explains why reported forecasts and investment decisions are internally inconsistent.
- The mechanism holds for TSE students and Prolific subjects.

8 Conclusion

8.1 Core conclusion

The paper shows that information changes both beliefs and the degree to which beliefs are used in portfolio decisions.

8.2 Economic intuition

Forecasts based on predictive signals are treated as more reliable; extrapolative forecasts are treated as more cognitively uncertain and thus have weaker decision impact.

8.3 Methodological implication

Observed portfolios cannot be interpreted without observing forecasts and perceived information sets. Otherwise, one may incorrectly infer generic underreaction or weak extrapolation.

8.4 Market implication

The results help reconcile household-investor evidence with professional-investor evidence and suggest that demand elasticities depend on the information environment.

8.5 Limitations

The experiment abstracts from transactions costs, equilibrium price feedback, and multi-asset portfolios, but isolates the key mechanism.

8.6 Takeaway

Return predictability, expectations, and investment are jointly determined; information changes not only what investors believe but also how they act on those beliefs.